

1 Fama and French (1993) replicate 결과

Table 1: Comparison of Fama and French (1993) Table 1 and Replicated Results

Size quintile	Original					Replicate				
	Low	2	3	4	High	Low	2	3	4	High
<i>Panel A: Average of annual averages of firm size</i>										
Small	20.6	20.8	20.2	19.4	15.1	21.6	21.7	21.2	20.0	15.6
2	89.7	89.3	89.3	89.9	88.5	97.1	96.6	96.5	95.3	93.8
3	209.3	211.9	210.8	214.8	210.7	228.0	230.0	232.0	232.0	228.0
4	535.1	537.4	545.4	551.6	538.7	569.0	577.0	587.0	591.0	581.0
Big	3583.7	2885.8	2819.5	2700.5	2337.9	3707.0	2978.0	2907.0	2745.0	2433.0
<i>Panel B: Average of annual BE/ME ratios for portfolio</i>										
Small	0.30	0.62	0.84	1.09	1.80	0.298	0.634	0.889	1.190	2.060
2	0.31	0.60	0.83	1.09	1.71	0.318	0.632	0.890	1.180	1.950
3	0.31	0.60	0.84	1.08	1.66	0.323	0.632	0.888	1.180	1.880
4	0.31	0.61	0.84	1.09	1.67	0.325	0.634	0.885	1.170	1.810
Big	0.29	0.59	0.83	1.08	1.56	0.311	0.624	0.875	1.170	1.750
<i>Panel C: Average of annual number of firms in portfolio</i>										
Small	428.0	276.6	263.8	291.5	512.7	441.0	280.0	264.0	294.0	522.0
2	121.6	94.0	86.7	79.8	71.3	123.0	90.8	89.2	82.7	70.9
3	102.7	78.3	73.0	64.5	45.9	98.2	72.9	70.9	61.6	44.9
4	90.1	68.9	60.7	53.1	33.4	82.6	64.3	56.7	50.2	33.0
Big	93.6	63.7	52.7	44.0	23.6	89.6	61.0	49.4	42.3	23.8

Table 2: Comparison of Fama and French (1993) Table 2 and Replicated Results

Size quintile	Original					Replicate				
	Low	2	3	4	High	Low	2	3	4	High
<i>Panel A: Means</i>										
Small	0.39	0.70	0.79	0.88	1.01	0.33	0.70	0.77	0.89	1.01
2	0.44	0.71	0.85	0.84	1.02	0.54	0.72	0.84	0.89	1.03
3	0.43	0.66	0.68	0.81	0.97	0.43	0.66	0.67	0.87	0.98
4	0.48	0.35	0.57	0.77	1.05	0.50	0.35	0.62	0.78	0.98
Big	0.40	0.36	0.32	0.56	0.59	0.39	0.36	0.34	0.55	0.52
<i>Panel B: Standard deviations</i>										
Small	7.76	6.84	6.29	5.99	6.27	7.77	6.77	6.24	5.94	6.25
2	7.28	6.42	5.85	5.33	6.06	7.37	6.32	5.76	5.28	6.04
3	6.71	5.71	5.27	4.92	5.69	6.63	5.68	5.19	4.95	5.71
4	5.97	5.44	5.03	4.95	5.75	5.93	5.40	5.06	4.94	5.66
Big	4.95	4.70	4.38	4.27	4.85	4.91	4.69	4.38	4.33	4.76
<i>Panel C: t-statistics for means</i>										
Small	0.93	1.88	2.33	2.73	2.97	0.79	1.93	2.31	2.81	3.02
2	1.11	2.05	2.69	2.91	3.11	1.38	2.14	2.73	3.14	3.19
3	1.18	2.12	2.39	3.04	3.15	1.20	2.18	2.41	3.30	3.21
4	1.49	1.19	2.08	2.88	3.36	1.56	1.22	2.29	2.93	3.22
Big	1.50	1.42	1.34	2.43	2.26	1.49	1.41	1.43	2.38	2.04

Table 3: Comparison of Fama and French (1993) Table 4 and Replicated Results

Size quintile	Original					Replicate				
	Low	2	3	4	High	Low	2	3	4	High
<i>Panel A: Market beta, b</i>										
Small	1.40	1.26	1.14	1.06	1.08	1.40	1.25	1.14	1.08	1.10
2	1.42	1.25	1.12	1.02	1.13	1.44	1.24	1.11	1.02	1.13
3	1.36	1.15	1.04	0.96	1.08	1.34	1.15	1.03	0.98	1.08
4	1.24	1.14	1.03	0.98	1.10	1.23	1.13	1.03	0.98	1.09
Big	1.03	0.99	0.89	0.84	0.89	1.01	0.98	0.87	0.85	0.86
<i>Panel B: $t(b)$</i>										
Small	26.33	28.12	27.01	25.03	23.01	27.29	29.27	28.58	27.44	24.94
2	35.76	35.56	33.12	33.14	29.04	37.03	37.07	35.05	35.47	30.87
3	42.98	42.52	37.50	35.81	31.16	45.61	44.77	40.13	39.12	31.84
4	51.67	55.12	46.96	37.00	32.76	53.53	58.47	48.31	39.94	33.91
Big	51.92	61.51	43.03	35.96	27.75	50.42	60.05	40.79	37.36	27.00
<i>Panel C: R^2</i>										
Small	0.67	0.70	0.68	0.65	0.61	0.68	0.71	0.70	0.69	0.64
2	0.79	0.79	0.76	0.76	0.71	0.80	0.80	0.78	0.78	0.73
3	0.84	0.84	0.80	0.79	0.74	0.86	0.85	0.82	0.82	0.75
4	0.89	0.90	0.87	0.80	0.76	0.89	0.91	0.87	0.82	0.77
Big	0.89	0.92	0.84	0.79	0.69	0.88	0.91	0.83	0.80	0.68
<i>Panel D: Residual standard error, $s(e)$</i>										
Small	4.46	3.76	3.55	3.56	3.92	4.38	3.64	3.41	3.34	3.74
2	3.34	2.96	2.85	2.59	3.25	3.31	2.84	2.70	2.46	3.12
3	2.65	2.28	2.33	2.26	2.90	2.51	2.18	2.19	2.13	2.89
4	2.01	1.73	1.84	2.21	2.83	1.95	1.64	1.82	2.09	2.73
Big	1.66	1.35	1.73	1.95	2.69	1.70	1.39	1.82	1.93	2.71

Table 4: Comparison of Fama and French (1993) Table 5 and Replicated Results

Size quintile	Original					Replicate				
	Low	2	3	4	High	Low	2	3	4	High
<i>Panel A: SMB loading, s</i>										
Small	1.93	1.73	1.63	1.59	1.67	1.96	1.74	1.66	1.60	1.70
2	1.52	1.46	1.35	1.18	1.40	1.60	1.48	1.38	1.20	1.43
3	1.28	1.12	1.05	0.93	1.16	1.29	1.17	1.07	0.96	1.22
4	0.86	0.82	0.77	0.72	0.95	0.90	0.84	0.80	0.79	0.97
Big	0.28	0.35	0.22	0.29	0.44	0.32	0.36	0.24	0.34	0.48
<i>Panel B: $t(s)$</i>										
Small	22.52	21.38	21.88	22.30	22.16	22.16	21.35	22.39	22.26	22.56
2	17.23	17.68	17.08	15.47	16.42	17.52	18.08	17.85	15.70	16.87
3	14.43	13.89	13.42	12.13	13.45	14.67	14.50	13.79	12.26	14.21
4	10.16	9.64	9.29	8.57	10.02	10.65	9.86	9.60	9.47	10.36
Big	3.70	4.39	2.79	3.69	5.02	4.18	4.47	2.99	4.21	5.60
<i>Panel C: HML loading, h</i>										
Small	-0.95	-0.57	-0.35	-0.18	0.01	-0.91	-0.50	-0.30	-0.14	0.05
2	-1.23	-0.66	-0.38	-0.16	0.00	-1.11	-0.61	-0.33	-0.11	0.05
3	-1.09	-0.65	-0.31	-0.11	0.01	-1.08	-0.59	-0.27	-0.07	0.06
4	-1.11	-0.65	-0.36	-0.11	-0.01	-1.09	-0.62	-0.32	-0.06	0.06
Big	-1.07	-0.65	-0.42	-0.06	0.08	-1.01	-0.62	-0.41	-0.05	0.18
<i>Panel D: $t(h)$</i>										
Small	-9.72	-6.19	-4.10	-2.20	0.16	-9.07	-5.39	-3.59	-1.76	0.55
2	-12.25	-7.02	-4.20	-1.82	0.05	-10.80	-6.57	-3.80	-1.24	0.54
3	-10.84	-7.07	-3.43	-1.23	0.12	-10.88	-6.49	-3.09	-0.76	0.63
4	-11.43	-6.69	-3.80	-1.12	-0.09	-11.45	-6.44	-3.44	-0.65	0.61
Big	-12.46	-7.07	-4.64	-0.66	0.81	-11.66	-6.80	-4.53	-0.54	1.90
<i>Panel E: R^2</i>										
Small	0.65	0.60	0.60	0.60	0.59	0.64	0.60	0.61	0.60	0.60
2	0.59	0.53	0.49	0.42	0.44	0.57	0.53	0.50	0.42	0.45
3	0.51	0.43	0.37	0.31	0.35	0.51	0.44	0.38	0.31	0.37
4	0.43	0.30	0.24	0.18	0.23	0.44	0.30	0.24	0.21	0.24
Big	0.34	0.18	0.08	0.04	0.06	0.32	0.17	0.09	0.05	0.09
<i>Panel F: Residual standard error, $s(e)$</i>										
Small	4.57	4.31	3.98	3.79	4.01	4.67	4.32	3.92	3.78	3.98
2	4.68	4.41	4.20	4.06	4.53	4.84	4.33	4.07	4.02	4.48
3	4.71	4.31	4.19	4.10	4.60	4.64	4.26	4.11	4.13	4.55
4	4.53	4.55	4.40	4.48	5.06	4.46	4.51	4.42	4.40	4.96
Big	4.02	4.27	4.20	4.19	4.69	4.06	4.28	4.20	4.23	4.56

Table 5: Comparison of Fama and French (1993) Table 6 and Replicated Results

Size quintile	Original					Replicate				
	Low	2	3	4	High	Low	2	3	4	High
<i>Panel A: Market loading, b</i>										
Small	1.04	1.02	0.95	0.91	0.96	1.03	0.99	0.93	0.90	0.95
2	1.11	1.06	1.00	0.97	1.09	1.13	1.03	0.97	0.96	1.07
3	1.12	1.02	0.98	0.97	1.09	1.10	1.01	0.96	0.98	1.07
4	1.07	1.08	1.04	1.05	1.18	1.06	1.07	1.03	1.04	1.15
Big	0.96	1.02	0.98	0.99	1.06	0.97	1.02	0.97	1.00	1.03
<i>Panel B: $t(b)$</i>										
Small	39.37	51.80	60.44	59.73	57.89	33.98	43.88	54.94	57.68	56.78
2	52.49	61.18	55.88	61.54	65.52	47.18	54.15	55.20	59.91	62.57
3	56.88	53.17	50.78	54.38	52.52	56.57	52.94	50.30	57.24	51.18
4	53.94	53.51	51.21	47.09	46.10	56.28	53.54	49.66	51.84	47.22
Big	60.93	56.76	46.57	53.87	38.61	58.36	58.21	44.45	54.51	39.32
<i>Panel C: SMB loading, s</i>										
Small	1.46	1.26	1.19	1.17	1.23	1.41	1.21	1.16	1.11	1.19
2	1.00	0.98	0.88	0.73	0.89	1.00	0.93	0.86	0.68	0.85
3	0.76	0.65	0.60	0.48	0.66	0.69	0.63	0.55	0.43	0.65
4	0.37	0.33	0.29	0.24	0.41	0.33	0.27	0.25	0.23	0.35
Big	-0.17	-0.12	-0.23	-0.17	-0.05	-0.20	-0.19	-0.28	-0.20	-0.07
<i>Panel D: $t(s)$</i>										
Small	37.92	44.11	52.03	52.85	50.97	30.96	35.43	45.61	47.16	47.42
2	32.73	38.79	34.03	31.66	36.78	27.75	32.63	32.50	28.04	32.95
3	26.40	23.39	21.23	18.62	21.99	23.65	21.95	19.17	16.60	20.58
4	12.73	11.11	9.81	7.38	11.01	11.59	8.92	7.81	7.65	9.51
Big	-7.18	-4.51	-7.58	-6.27	-1.18	-8.01	-7.10	-8.67	-7.33	-1.84
<i>Panel E: HML loading, h</i>										
Small	-0.29	0.08	0.26	0.40	0.62	-0.30	0.09	0.25	0.39	0.61
2	-0.52	0.01	0.26	0.46	0.70	-0.45	0.00	0.24	0.46	0.69
3	-0.38	0.00	0.32	0.51	0.68	-0.42	0.01	0.30	0.52	0.70
4	-0.42	0.04	0.30	0.56	0.74	-0.46	0.01	0.29	0.56	0.75
Big	-0.46	0.00	0.21	0.57	0.76	-0.44	-0.02	0.17	0.55	0.80
<i>Panel F: $t(h)$</i>										
Small	-6.47	2.35	9.66	15.53	22.24	-5.79	2.44	8.78	14.86	21.59
2	-14.57	0.41	8.56	17.24	24.80	-11.06	0.03	8.20	17.08	23.73
3	-11.26	0.05	9.75	16.88	19.39	-12.80	0.21	9.30	17.76	19.69
4	-12.51	1.04	8.83	14.84	17.09	-14.48	0.38	8.22	16.39	18.12
Big	-17.03	0.09	5.80	18.34	16.24	-15.52	-0.52	4.58	17.55	17.95

Table 6: Comparison of Fama and French (1993) Table 9a and Replicated Results

Size quintile	Original					Replicate				
	Low	2	3	4	High	Low	2	3	4	High
<i>Panel A: Regression (ii), intercepts</i>										
Small	-0.22	0.15	0.30	0.42	0.54	-0.28	0.16	0.28	0.43	0.54
2	-0.18	0.17	0.36	0.39	0.53	-0.08	0.19	0.36	0.45	0.55
3	-0.16	0.15	0.23	0.39	0.50	-0.15	0.17	0.23	0.45	0.52
4	-0.05	-0.14	0.12	0.35	0.57	-0.03	-0.13	0.17	0.35	0.51
Big	-0.04	-0.07	-0.07	0.20	0.21	-0.04	-0.07	-0.04	0.19	0.15
<i>Panel B: Regression (ii), t-statistics</i>										
Small	-0.90	0.73	1.54	2.19	2.53	-1.17	0.81	1.52	2.38	2.67
2	-1.00	1.05	2.35	2.79	3.01	-0.43	1.24	2.50	3.38	3.24
3	-1.12	1.25	1.82	3.20	3.19	-1.14	1.44	1.92	3.95	3.33
4	-0.50	-1.50	1.20	2.91	3.71	-0.30	-1.50	1.78	3.13	3.47
Big	-0.49	-0.95	-0.70	1.89	1.41	-0.46	-0.91	-0.42	1.79	1.02
<i>Panel C: Regression (iii), intercepts</i>										
Small	0.24	0.46	0.49	0.53	0.55	0.20	0.46	0.47	0.54	0.56
2	0.52	0.58	0.64	0.58	0.64	0.59	0.59	0.63	0.63	0.65
3	0.52	0.61	0.52	0.60	0.66	0.53	0.61	0.51	0.66	0.65
4	0.69	0.39	0.50	0.62	0.79	0.71	0.39	0.55	0.60	0.70
Big	0.76	0.52	0.43	0.51	0.44	0.72	0.51	0.44	0.49	0.32
<i>Panel D: Regression (iii), t-statistics</i>										
Small	0.97	1.92	2.24	2.52	2.49	0.77	1.94	2.20	2.64	2.58
2	2.00	2.40	2.76	2.61	2.56	2.22	2.50	2.82	2.86	2.65
3	2.00	2.58	2.25	2.66	2.61	2.11	2.60	2.26	2.92	2.61
4	2.78	1.55	2.07	2.51	2.85	2.91	1.58	2.27	2.49	2.60
Big	3.41	2.23	1.84	2.20	1.70	3.25	2.20	1.91	2.11	1.29
<i>Panel E: Regression (iv), intercepts</i>										
Small	-0.34	-0.12	-0.05	0.01	0.00	-0.35	-0.08	-0.03	0.06	0.05
2	-0.11	-0.01	0.08	0.03	0.02	-0.01	0.04	0.11	0.11	0.07
3	-0.11	0.04	-0.04	0.05	0.05	-0.06	0.07	-0.01	0.13	0.07
4	0.09	-0.22	-0.08	0.03	0.13	0.14	-0.18	-0.01	0.04	0.09
Big	0.21	-0.05	-0.13	-0.05	-0.16	0.20	-0.03	-0.08	-0.05	-0.23
<i>Panel F: Regression (iv), t-statistics</i>										
Small	-3.16	-1.47	-0.73	0.22	0.14	-2.85	-0.82	-0.38	0.98	0.80
2	-1.24	-0.20	1.04	0.51	0.34	-0.15	0.56	1.54	1.74	1.06
3	-1.42	0.47	-0.47	0.71	0.56	-0.70	0.87	-0.10	1.89	0.88
4	1.07	-2.65	-0.99	0.33	1.24	1.85	-2.22	-0.06	0.54	0.88
Big	3.27	-0.67	-1.46	-0.69	-1.48	2.98	-0.43	-0.89	-0.65	-2.14